

Evidence-Based Vulnerability Assessment and Red Teaming Methodologies for Large Language Models

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Abstract—As Large Language Models (LLMs) are increasingly integrated into critical infrastructure, identifying systemic vulnerabilities through rigorous testing is paramount. This paper presents an evidence-based audit and red teaming framework designed to expose harmful content generation, latent biases, and security bypasses. By synthesizing empirical data from recent safety benchmarks—including the Bias Benchmark for QA (BBQ) and Greedy Coordinate Gradient (GCG) attack methodologies—this study identifies key technical failure points in state-of-the-art models such as Llama 3 and GPT-4o.

The research proposes a standardized, four-stage red teaming protocol—comprising reconnaissance, automated probing, semantic obfuscation, and metric-driven reporting—to identify risks before deployment. Detailed analysis reveals significant vulnerabilities, including a 100% failure rate in specific indirect prompt injection tests and high susceptibility to adversarial suffixes. Findings indicate that while current safety guardrails like Reinforcement Learning from Human Feedback (RLHF) are improving, models remain vulnerable to complex adversarial prompt injections, linguistic obfuscation, and subtle stereotype amplification. Specifically, empirical evidence demonstrates that automated scans can achieve up to an 80% success rate in bypassing safety filters under certain stress conditions. This assessment provides a technical foundation for developing more resilient AI safety architectures and argues for the necessity of continuous, automated red teaming to keep pace with evolving adversarial tactics.

Keywords—Large Language Models (LLMs), Red Teaming, Vulnerability, Jailbreaking, Llama, Bias

I. INTRODUCTION

The rapid deployment of generative artificial intelligence has outpaced the development of standardized security protocols, leading to significant concerns regarding model reliability and ethical alignment [1]. AI Red Teaming, a methodology adapted from traditional cybersecurity, has emerged as the primary defensive strategy for simulating adversarial attacks to discover hidden vulnerabilities. Unlike standard software testing, red teaming for LLMs must address non-deterministic outputs where harmful content or social biases may only emerge under specific stress conditions. This report aims to bridge the gap between theoretical risk and technical proof. By analyzing recent red teaming literature and model safety cards—specifically Meta’s Llama 3 and OpenAI’s GPT-

4o—this study documents the specific triggers that lead to safety failures. The objective is to provide a reproducible auditing framework that enables proactive risk mitigation.

II. LITERATURE REVIEW

Recent research identifies certain critical failure modes in aligned LLMs:

1) *Adversarial Jailbreaking*: Techniques like GCG demonstrate that suffix-based optimization can bypass safety filters with high success rates.

2) *Algorithm Bias*: Studies utilizing the BBQ benchmark highlight that models often revert to harmful stereotype when queries with ambiguous prompts, particularly concerning gender and ethnicity.

3) *Linguistic Obfuscation*: Standard safety filters are frequently less effective against inputs encoded in Base64 or rare languages, as shown in multiple vulnerability reports.

III. METHODOLOGY

This assessment utilizes a Secondary Research-Based Audit framework. The methodology involves:

1) *Selection Criteria*: Analysis of peer-reviewed papers from 2024-2025 and official model system cards.

2) *Vulnerability Classification*: Categorization of risks into Jailbreaking, Social Bias, and Content Safety.

3) *Benchmark Analysis*: Evaluating model performance against standardised datasets such as ToxiGen and CyberSecEval.

IV. EVIDENCED BASED FINDINGS

Based on the analyzed data, the following vulnerabilities represent the current state-of-play in LLM risks.

Vulnerability Type	Technical Proof / Stress Test	Documented Evidence / Metric
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Jailbreaking	GCG Adversarial Suffixes	Llama 3 models remain vulnerable to optimized adversarial suffixes with a minimum ASR of 0.88.
Prompt Injection	Indirect Injection (ADD/IIO)	Llama 3.3 70B shows a complete failure (100/100 risk score) in indirect instruction override tests.
Social Bias	BBQ Benchmark Analysis	Research shows 64% of interactions in specific scenarios result in bias assessments due to anchoring and availability bias.
Harmful Content	Toxic Trigger Probs.	Garak-based automated scans revealed that Pliny-style injection achieved up to 80% success in GPT-4o

V. PROPOSED RED TIMING PROTOCOL

To mitigate the identified risks, this report proposes a four-stage Evidence-Based Red Timing Protocol:

1) *Reconnaissance*: Identify the model’s training data limits and known “blind spots” (e.g., low-resource languages)

2) *Automated Probing*: Deploy open source scanners like Garak or Microsoft Pyrit to run standardise attack suites.

3) *Semantic Obfuscation Testing*: Manually test the model using ciphers (ROT13) and roleplay (DAN) to evaluate instruction-following robustness.

4) *Metric-Driven Reporting*: Quantify failure rates using Attack Success Rate (ASR) and Toxicity Scores from the Perspective API.

VI. CONCLUSION

The evidence confirms that while safety alignment (e.g., RLHF) has significantly reduced direct harmful outputs, models remain highly susceptible to sophisticated adversarial techniques. The 100% failure rate of some models against indirect prompt injections underscore a critical gap in current defense architectures. Future auditing must move towards automated, continuous red teaming to keep pace with evolving attack vectors.

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This document was drafted with the assistance of Google’s AI Overviews, Gemini 3.1 Pro. Under strick supervision, technical review, and final approval of Daniel David, AI Safety & Governance Analyst – Global Multilateral AI Committee. We acknowledge the ethical guidelines of the Institute of Electrical and Electronics Engineers on AI authorship and the foundation principles of data protection and privacy at the international and national levels.